



PRESIDENCY UNIVERSITY

Private University Estd. in Karnataka State by Act No. 41 of 2013

Itgalpura, Rajankunte, Yelahanka, Bengaluru – 560064

STUDENT ACADEMIC PERFORMANCE ANALYSIS

USING SQL AND DATA ANALYTICS

CSS7000 INTERNSHIP REPORT

Submitted by

SINDHU HOSAGOUDAR — 20241CIT0179

Under the guidance of,

Dr. Ramesh Sengodan

COMPUTER SCIENCE AND ENGINEERING

(Internet of things)



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'Y Internship

PRESIDENCY SCHOOL OF ARTIFICIAL INTELLIGENCE & ADVANCED COMPUTING

BONAFIDE CERTIFICATE

Certified that this report “**STUDENT ACADEMIC PERFORMANCE ANALYSIS USING SQL AND DATA ANALYTICS**” is a bonafide work of **Sindhu Hosagoudar (20241CIT0179)**, who has successfully carried out the internship work and submitted the report for partial fulfilment of the requirements for the award of the degree of BACHELOR OF TECHNOLOGY in PRESIDENCY SCHOOL OF ARTIFICIAL INTELLIGENCE & ADVANCED COMPUTING ,Internet Of Things during 2026-2027

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DECLARATION

I, Sindhu Hosagoudar, a student of Third year B.Tech in Computer Science and Engineering, (INTERNET OF THINGS) bearing roll number 20241CIT0179, at Presidency University, Bengaluru, hereby declare that the internship work titled "Student Academic Performance Analysis Using SQL and Data Analytics", carried out at ZWAPY (Company ID: ZWAPY), has been independently completed by me under the guidance of Dr. Ramesh Sengodan and has been submitted in partial fulfilment of the requirements for the award of the degree of Bachelor of Technology during the academic year 2026-2027. I further declare that the matter embodied in this report has not been submitted previously, in part or in full, for the award of any other degree or diploma to any other institution.

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INTERNSHIP COMPLETION CERTIFICATE



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ABSTRACT

Educational institutions generate large volumes of academic and behavioral data every semester attendance records, assignment scores, internal assessments, and final examination marks yet this data is frequently underutilized for identifying students who may need additional academic support. This internship project, carried out at RIVOQUIX, addresses this gap by building a structured, data-driven pipeline for analyzing student academic performance using SQL and Python-based data analytics.

The project uses a student dataset containing demographic details (gender, age, department, semester), behavioral indicators (attendance percentage, daily study hours), and academic outcomes (assignment scores, internal marks, final examination marks, overall grade, and pass/fail status). The raw dataset was first cleaned using Python (Pandas and NumPy) to handle missing values, remove duplicate records, and correct inconsistent entries. A relational table was then created in SQL to store the cleaned data, and a series of SQL queries were written to filter records, compute averages, minimums and maximums, group students by department, semester and grade, calculate pass percentages, and identify top-performing and at-risk students.

On the analytics side, exploratory data analysis (EDA) was performed using Pandas, NumPy, Matplotlib and Seaborn to study statistical summaries and correlations between behavioral factors and academic outcomes. Visualizations including scatter plots, bar charts, pie charts, and a correlation heatmap were used to communicate patterns clearly. The analysis shows a strong positive relationship between attendance, daily study hours, and overall academic performance, with study hours and internal assessment marks emerging as the strongest predictors of final examination performance among the factors studied.

The project deliberately keeps its scope within data analytics and SQL, in line with the internship requirement, and does not extend into machine learning model building. The resulting insights

— such as the identification of low-attendance, low-study-hour student segments that are strongly associated with lower academic outcomes — provide a practical basis on which an institution could design early-warning and academic support mechanisms for at-risk students.

Keywords: Student Performance Analysis, SQL, Data Analytics, Exploratory Data Analysis, Attendance, Academic Support, Python, Pandas, Data Visualization.

1. Introduction

1.1 About the Internship

This report documents the work carried out during an internship focused on applying SQL and data analytics techniques to a real-world style problem: understanding and analyzing student academic performance. The internship required building an end-to-end data workflow starting from raw, imperfect data, moving through cleaning and structured storage in SQL, and finishing with statistical analysis and visualization rather than building a predictive machine learning model. The emphasis throughout was on descriptive and diagnostic analytics: understanding what the data shows and why certain patterns occur, using SQL for structured querying and Python for analysis and visualization.

1.2 About the Company — RIVOQUIX

The internship was carried out at ZWAPY (Company ID: RIVOQUIX). As part of the internship, the intern was assigned a data analytics problem statement centered on educational data, with the objective of demonstrating practical proficiency in SQL-based data management and Python-based data analytics. The work was periodically reviewed by the assigned industry reviewer, Mr. Tanveer Ahmed, to ensure the analysis remained grounded, methodologically sound, and aligned with the stated scope of the assignment.

1.3 Problem Statement

Academic institutions routinely collect data on student attendance, assignments, internal assessments, and final examinations, but this data is often stored across disconnected spreadsheets or systems and is rarely analyzed systematically. As a result, students who are gradually falling behind — due to low attendance, insufficient study time, or weak performance in continuous assessments — are often identified too late, if at all. The problem addressed in this project is: given a structured dataset of student academic and behavioral records, how can SQL and data analytics be used to (a) organize and query this data efficiently, and (b) identify the factors most strongly associated with academic success or failure, so that institutions can proactively identify students who may need additional support.

2. Objectives of the Project

The internship project was guided by the following specific objectives:

- To design and populate a relational SQL table capable of storing student academic and behavioral data in a structured, queryable form.
- To write SQL queries for filtering student records, computing aggregate statistics (average, minimum, maximum), grouping students by department, semester and grade, and calculating pass percentages.
- To identify top-performing and at-risk (low-performing) students using SQL-based ranking and filtering queries.
- To clean and preprocess the raw dataset using Python (Pandas, NumPy) — handling missing values, duplicate records, and inconsistent entries.
- To perform exploratory data analysis (EDA) to understand statistical distributions and relationships within the data.
- To examine the relationship between attendance, study hours, and academic performance through correlation analysis.
- To visualize key patterns using Matplotlib and Seaborn — including attendance-vs-marks trends, grade distribution, department-wise comparisons, and a correlation heatmap.
- To summarize findings into actionable insights that could help an institution identify students needing academic support.

Consistent with the scope of the internship, the project deliberately excludes machine learning model building (e.g., classification or regression models to predict grades). The focus remains entirely on SQL-based data management and data analytics / EDA techniques, which is the core skill set the internship was designed to develop.

3. Literature Review / Existing Approaches

Student performance analysis is a well-studied application area within educational data analytics. Institutions traditionally rely on simple spreadsheet-based reporting to track marks and attendance, which is adequate for record-keeping but limited for identifying underlying patterns across large cohorts. A more structured approach — storing student records in a relational database and using SQL for querying — allows for consistent, repeatable, and scalable analysis across departments, semesters, and academic years.

Beyond structured querying, exploratory data analysis (EDA) techniques from the data analytics domain are commonly used to study relationships between behavioral factors (such as attendance and study time) and academic outcomes. Descriptive statistics, grouping, and correlation analysis are widely used first steps before any advanced modeling is attempted, as they help validate whether the data actually supports the assumptions typically made about student performance (for example, the common assumption that attendance is positively related to marks).

This project follows this analytics-first approach: it does not attempt to build a predictive model, but instead focuses on establishing, through SQL queries and statistical/visual analysis, which behavioral and academic factors are meaningfully associated with overall performance in the dataset used. This is consistent with the stated internship requirement of keeping the project within the SQL and Data Analytics domain rather than extending into advanced machine learning.

4. Tools and Technologies Used

The following tools and technologies were used to carry out the internship project:

Tool / Technology	Purpose in this Project
Python 3	Core scripting language used for data cleaning, analysis, and chart generation
Pandas	Data loading, cleaning, grouping, and aggregation
NumPy	Numerical operations and array-based computations
Matplotlib	Base plotting library for charts
Seaborn	Statistical visualizations (scatter plots, heatmaps, bar charts) built on Matplotlib
SQL (SQLite/MySQL syntax)	Table creation, filtering, aggregation, grouping and ranking queries
Jupyter / Script-based workflow	Iterative data exploration and reproducible analysis scripts

SQL was used as the primary language for structured data storage and querying, simulating how an institution's student information system might store records in a relational database (e.g., MySQL). Python's data analytics stack (Pandas, NumPy, Matplotlib, Seaborn) was used for cleaning, statistical analysis, and visualization — the two together form the complete SQL + Data Analytics workflow required for this internship.

5. System Requirements

5.1 Hardware Requirements

Component	Requirement
Processor	Intel Core i5 (or equivalent) and above
RAM	8 GB minimum

5.2 Software Requirements

Component	Requirement
Operating System	Windows 10 / 11, macOS, or Linux
Programming Language	Python 3.9 or higher
Database	MySQL 8.x / SQLite 3
Python Libraries	pandas, numpy, matplotlib, seaborn
IDE / Tools	Jupyter Notebook / VS Code, MySQL Workbench (or equivalent SQL client)

6. Dataset Description

The dataset used for this project is a structured student academic and behavioral dataset containing 480 cleaned student records after preprocessing (490 raw records before cleaning). It combines demographic information, behavioral indicators, and academic outcome measures for students across five departments (CSE, ISE, ECE, AIML, CIVIL) and four semesters (3 to 6).

6.1 Data Features

Column	Description
Student_ID	Unique identifier assigned to each student
Gender	Male / Female
Age	Age of the student in years
Department	Academic department (CSE, ISE, ECE, AIML, CIVIL)
Semester	Current semester of the student (3 to 6)
Attendance_Percentage	Percentage of classes attended
Study_Hours_per_Day	Average number of hours studied per day
Assignment_Score	Score obtained in assignments (out of 100)
Internal_Marks	Internal assessment marks (out of 50)
Final_Exam_Marks	Marks obtained in the final examination (out of 100)
Total_Marks	Sum of Internal_Marks and Final_Exam_Marks (out of 150)
Overall_Percentage	Total_Marks converted to a percentage (out of 100)
Grade	Letter grade derived from Overall_Percentage (O, A+, A, B+, B, C, F)
Pass_Fail	Pass if Overall_Percentage $\geq$ 40, else Fail

6.2 Sample Records (after cleaning)

Student_ID	Gender	Dept	Sem	Attend.%	Study Hrs	Internal	Final	Overall%	Grade
STU1060	Female	CSE	4	73.0	2.7	36.7	42.5	52.8	B
STU1462	Female	CSE	6	72.4	6.2	38.1	67.4	70.3	A
STU1402	Female	AIML	6	75.9	2.1	32.6	35.1	45.1	C
STU1051	Male	ECE	6	74.2	5.7	30.7	47.0	51.8	B
STU1159	Female	ECE	5	72.5	1.1	28.5	19.6	32.1	F

Dataset size: 480 records (after removing 10 duplicate entries from 490 raw records). No missing values remain in the cleaned dataset (see Section 7.2 for the cleaning process).

7. Methodology

7.1 Overall Workflow

The project followed a structured, five-stage workflow that mirrors how such an analysis would typically be carried out within a real institutional or company setting:

- Stage 1 — Data Collection: Compilation of the raw student dataset containing demographic, behavioral and academic fields.
- Stage 2 — Data Cleaning & Preprocessing: Using Python (Pandas/NumPy) to remove duplicate records, handle missing values, and correct inconsistent entries.
- Stage 3 — Structured Storage (SQL): Loading the cleaned data into a SQL table to enable structured querying.
- Stage 4 — SQL-based Querying: Writing SQL queries for filtering, aggregation, grouping and ranking to answer specific academic questions.
- Stage 5 — Data Analytics & Visualization: Performing EDA, statistical summaries, correlation analysis, and visualization in Python to interpret and communicate the SQL-derived results.

7.2 Data Cleaning and Preprocessing

The raw dataset (490 records) contained realistic data-quality issues that needed to be resolved before analysis:

- Duplicate records: 10 duplicate Student_ID rows were identified and removed, keeping the first occurrence.
- Missing values: Attendance_Percentage (9 records), Study_Hours_per_Day (7 records), Assignment_Score (9 records) and Gender (4 records) contained missing entries. Numeric fields were imputed using the department-wise median to preserve realistic department-level distributions; Gender was imputed using the overall mode.
- Inconsistent text entries: Gender values such as 'male' and 'FEMALE' (inconsistent casing) were standardized to a consistent 'Male' / 'Female' format.

- Invalid values: A small number of negative Study_Hours_per_Day entries (a data-entry error, since study hours cannot be negative) were treated as missing and imputed.

After cleaning, the dataset contained 480 records with zero missing values across all columns, which was verified programmatically before proceeding to the SQL and analytics stages. Table 7.1 summarizes the before/after state of the cleaning process.

Metric	Before Cleaning	After Cleaning
Total Records	490	480
Duplicate Student IDs	10	0
Missing Attendance_Percentage	9	0
Missing Study_Hours_per_Day	7	0
Missing Assignment_Score	9	0
Missing Gender	4	0
Invalid (negative) Study Hours	2	0

8. SQL Component

This section presents the core SQL work carried out on the cleaned student dataset. All queries were executed against a table named `students`, structured to mirror how the data would be stored in a MySQL database, and all result values shown below were produced directly by running these queries against the actual cleaned dataset described in Chapter 6.

8.1 Table Creation

The following DDL statement was used to create the `students` table:

```
CREATE TABLE students (
  Student_ID TEXT PRIMARY KEY,
  Gender TEXT,
  Age INTEGER,
  Department TEXT,
  Semester INTEGER,
```

```

Attendance_Percentage REAL,
Study_Hours_per_Day REAL,
Assignment_Score REAL,
Internal_Marks REAL,
Final_Exam_Marks REAL,
Total_Marks REAL,
Overall_Percentage REAL,
Grade TEXT,
Pass_Fail TEXT
);

```

8.2 Filtering and Aggregate Queries

8.2.1 Overall summary statistics (MIN, MAX, AVG)

```

SELECT
  MIN(Overall_Percentage) AS Min_Percentage,
  MAX(Overall_Percentage) AS Max_Percentage,
  ROUND(AVG(Overall_Percentage),2) AS Avg_Percentage,
  ROUND(AVG(Attendance_Percentage),2) AS Avg_Attendance,
  ROUND(AVG(Study_Hours_per_Day),2) AS Avg_Study_Hours
FROM students;

```

Min %	Max %	Avg %	Avg Attendance	Avg Study Hrs
7.3	100	44.74	70.98	3.15

8.2.2 Attendance-band vs performance

```

SELECT
  CASE WHEN Attendance_Percentage < 75 THEN 'Below 75%' ELSE '75% and Above' END AS
Attendance_Band,
  ROUND(AVG(Overall_Percentage),2) AS Avg_Percentage,
  COUNT(*) AS Num_Students
FROM students
GROUP BY Attendance_Band;

```

Attendance Band	Avg Overall %	No. of Students
-----------------	---------------	-----------------

Attendance Band	Avg Overall %	No. of Students
75% and Above	64.06	170
Below 75%	34.14	310

8.2.3 Study-hour band vs performance

```

SELECT
  CASE
    WHEN Study_Hours_per_Day < 2 THEN '<2 hrs'
    WHEN Study_Hours_per_Day < 4 THEN '2-4 hrs'
    WHEN Study_Hours_per_Day < 6 THEN '4-6 hrs'
    ELSE '6+ hrs'
  END AS Study_Band,
  ROUND(AVG(Overall_Percentage),2) AS Avg_Percentage,
  COUNT(*) AS Num_Students
FROM students
GROUP BY Study_Band
ORDER BY Avg_Percentage;

```

Study Hour Band	Avg Overall %	No. of Students
<2 hrs	22.49	137
2-4 hrs	42.22	177
4-6 hrs	60.44	125
6+ hrs	82.07	41

8.3 Grouping and Pass Percentage Queries

8.3.1 Department-wise average performance

```
SELECT Department,
       ROUND(AVG(Overall_Percentage),2) AS Avg_Percentage,
       ROUND(AVG(Attendance_Percentage),2) AS Avg_Attendance,
       COUNT(*) AS Num_Students
FROM students
GROUP BY Department
ORDER BY Avg_Percentage DESC;
```

Department	Avg Overall %	Avg Attendance %	No. of Students
ISE	47.78	71.95	108
CSE	47.12	71.71	152
ECE	42.91	70.57	74
AIML	41.92	69.79	98
CIVIL	38.89	69.54	48

8.3.2 Semester-wise average performance

```
SELECT Semester,
       ROUND(AVG(Overall_Percentage),2) AS Avg_Percentage,
       COUNT(*) AS Num_Students
FROM students
GROUP BY Semester
ORDER BY Semester;
```

Semester	Avg Overall %	No. of Students
3	43.18	124
4	44.21	122

Semester	Avg Overall %	No. of Students
5	42.7	101
6	48.22	133

8.3.3 Grade distribution

```
SELECT Grade, COUNT(*) AS Num_Students,
       ROUND(100.0*COUNT(*)/(SELECT COUNT(*) FROM students),2) AS Percentage
FROM students
GROUP BY Grade
ORDER BY Num_Students DESC;
```

Grade	No. of Students	Percentage
F	203	42.29
C	90	18.75
B	70	14.58
B+	51	10.63
A	32	6.67
A+	20	4.17
O	14	2.92

8.3.4 Overall pass / fail percentage

```
SELECT Pass_Fail, COUNT(*) AS Num_Students,
       ROUND(100.0*COUNT(*)/(SELECT COUNT(*) FROM students),2) AS Percentage
FROM students
GROUP BY Pass_Fail;
```

Status	No. of Students	Percentage
Fail	203	42.29

Status	No. of Students	Percentage
Pass	277	57.71

8.3.5 Department-wise pass percentage

```
SELECT Department,
       ROUND(100.0*SUM(CASE WHEN Pass_Fail='Pass' THEN 1 ELSE 0 END)/COUNT(*),2) AS
Pass_Percentage,
       COUNT(*) AS Num_Students
FROM students
GROUP BY Department
ORDER BY Pass_Percentage DESC;
```

Department	Pass Percentage	No. of Students
ISE	64.81	108
CSE	60.53	152
ECE	58.11	74
AIML	54.08	98
CIVIL	39.58	48

8.4 Top Performers and At-Risk Students

8.4.1 Top 10 performing students

```
SELECT Student_ID, Department, Overall_Percentage, Grade
FROM students
ORDER BY Overall_Percentage DESC
LIMIT 10;
```

Student ID	Department	Overall %	Grade
STU1216	CIVIL	100	O
STU1369	ISE	100	O
STU1134	CSE	100	O
STU1357	ISE	99.7	O
STU1260	CIVIL	98.9	O
STU1305	ISE	97.5	O
STU1046	CSE	96.3	O
STU1297	AIML	95.5	O
STU1139	ISE	95	O
STU1427	AIML	93.6	O

8.4.2 At-risk students (Overall % below 45)

```
SELECT Student_ID, Department, Attendance_Percentage, Overall_Percentage, Grade
FROM students
WHERE Overall_Percentage < 45
ORDER BY Overall_Percentage ASC
LIMIT 10;
```

Student ID	Department	Attendance %	Overall %	Grade
------------	------------	--------------	-----------	-------

Student ID	Department	Attendance %	Overall %	Grade
STU1242	CSE	45.4	7.3	F
STU1037	CIVIL	51.6	7.4	F
STU1073	CSE	47.7	8.3	F
STU1255	ECE	55.8	8.4	F
STU1420	ISE	50.1	8.5	F
STU1382	AIML	50.9	8.6	F
STU1012	CSE	51.1	8.8	F
STU1150	AIML	65.7	9.3	F
STU1121	AIML	43.3	9.6	F
STU1114	AIML	57.4	9.7	F

Every one of the ten at-risk students identified above also has attendance below 66%, which is consistent with the attendance-band query in Section 8.2.2 showing markedly lower average performance for the below-75% attendance group.

9. Data Analytics Component

9.1 Statistical Summary

Descriptive statistics were computed on the cleaned dataset (480 records) using Pandas to understand the central tendency and spread of the key numerical variables:

Variable	Mean	Std. Dev.	Min	Median	Max
Attendance (%)	70.98	10.66	39.80	71.20	100.00
Study Hours / Day	3.14	1.88	0.00	3.10	7.60
Assignment Score	67.58	14.59	23.00	68.15	100.00
Internal Marks (/50)	31.01	7.95	10.90	31.10	50.00
Final Exam Marks (/100)	36.10	25.71	0.00	35.45	100.00
Overall Percentage	44.74	21.92	7.30	43.90	100.00

The relatively high standard deviation for Final_Exam_Marks and Overall_Percentage indicates a wide spread in performance across the student cohort, consistent with the grade distribution seen in Section 8.3.3, where a large share of students fall into either the top (O/A+) or bottom (F) grade bands.

9.2 Correlation Analysis

A Pearson correlation matrix was computed across all numeric academic and behavioral variables to quantify how strongly each factor is associated with Overall_Percentage. The full matrix is visualized as a heatmap in Chapter 10 (Figure 10.7); the correlations with Overall_Percentage specifically are summarized below:

Factor	Correlation with Overall %
Final_Exam_Marks	0.99
Internal_Marks	0.92

Factor	Correlation with Overall %
Study_Hours_per_Day	0.86
Assignment_Score	0.83
Attendance_Percentage	0.80
Age	-0.01

Final_Exam_Marks and Internal_Marks show the strongest correlation with Overall_Percentage, which is expected since these two components together make up the Overall_Percentage calculation. Among the independent behavioral factors, Study_Hours_per_Day (0.86) shows a slightly stronger association with performance than Attendance_Percentage (0.80), suggesting that, in this dataset, how much a student studies is marginally more predictive of outcomes than how often they attend class — though both are strongly and positively related to performance. Age shows a negligible correlation (-0.01), indicating that, within the 19-22 age range represented in this cohort, age itself is not a meaningful factor in academic performance.

10. Visualizations

This chapter presents the key visualizations generated from the cleaned dataset using Matplotlib and Seaborn. Each chart is derived directly from the same dataset and SQL query outputs discussed in Chapters 8 and 9, ensuring that the visual patterns are fully consistent with the reported figures.

10.1 Attendance vs Overall Marks

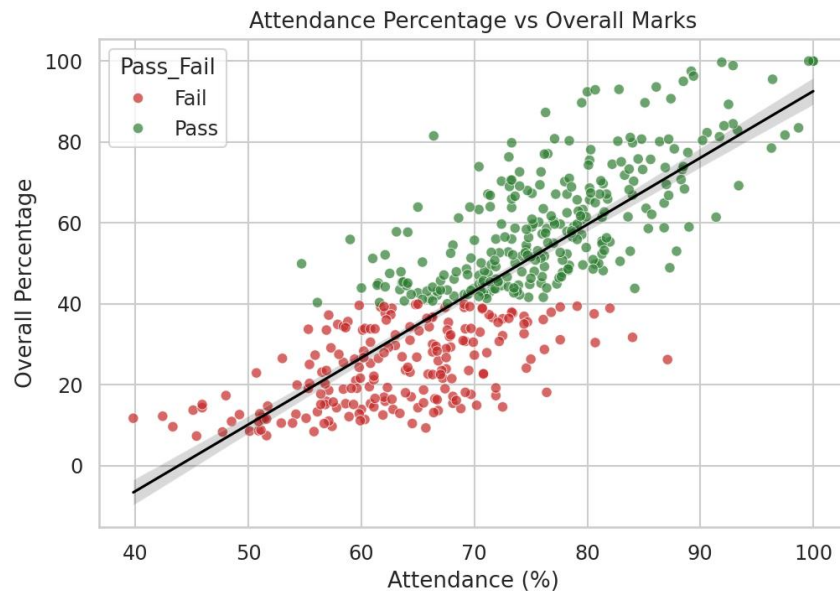


Figure 10.1 — Attendance Percentage vs Overall Marks, coloured by Pass/Fail status

The scatter plot shows a clear upward trend between attendance and overall marks, confirmed by the fitted trend line. Students with attendance below roughly 65% are overwhelmingly clustered in the 'Fail' category, while students above 85% attendance are almost entirely in the 'Pass' category.

10.2 Study Hours vs Overall Marks

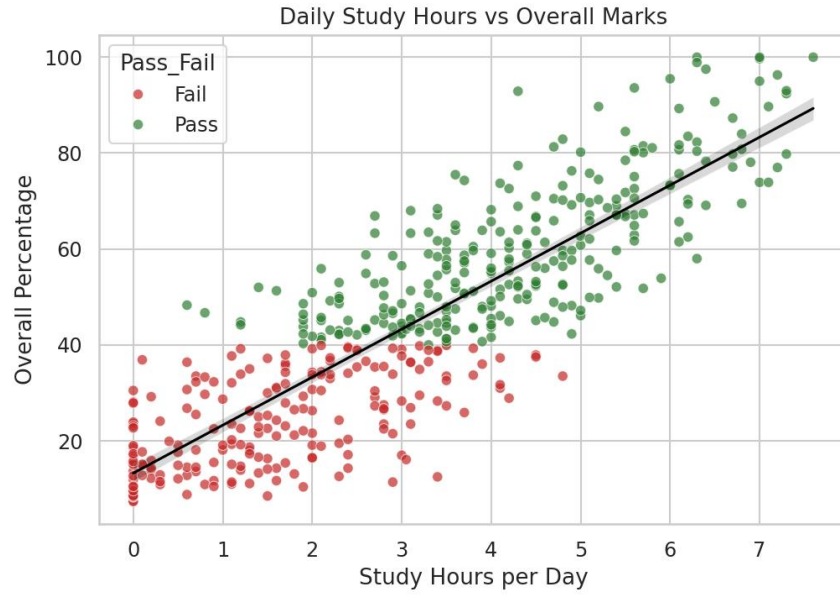


Figure 10.2 — Daily Study Hours vs Overall Marks, coloured by Pass/Fail status

Study hours show an even steeper positive relationship with overall marks than attendance, reinforcing the correlation values reported in Section 9.2. Students studying fewer than 2 hours a day average only 22.5% overall, compared to 82.1% for students studying more than 6 hours a day (see Section 8.2.3).

10.3 Grade Distribution

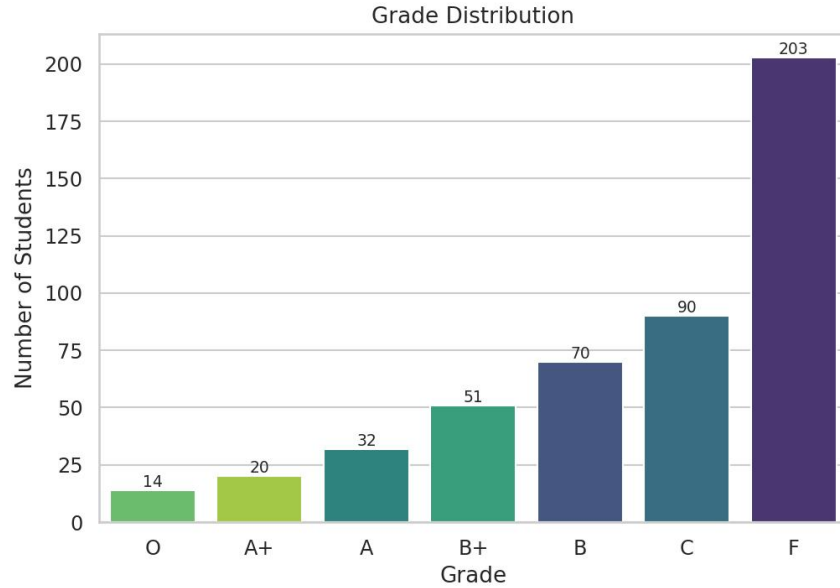


Figure 10.3 — Distribution of students across grade categories

The grade distribution is bimodal: a large cluster of students falls into the F grade (42.3%), while a substantial group achieves O/A+/A grades. Relatively fewer students fall into the middle bands (B, B+), suggesting that once a student falls behind on attendance and study consistency, the gap tends to compound rather than settle into a 'middling' outcome.

10.4 Department-wise Average Marks

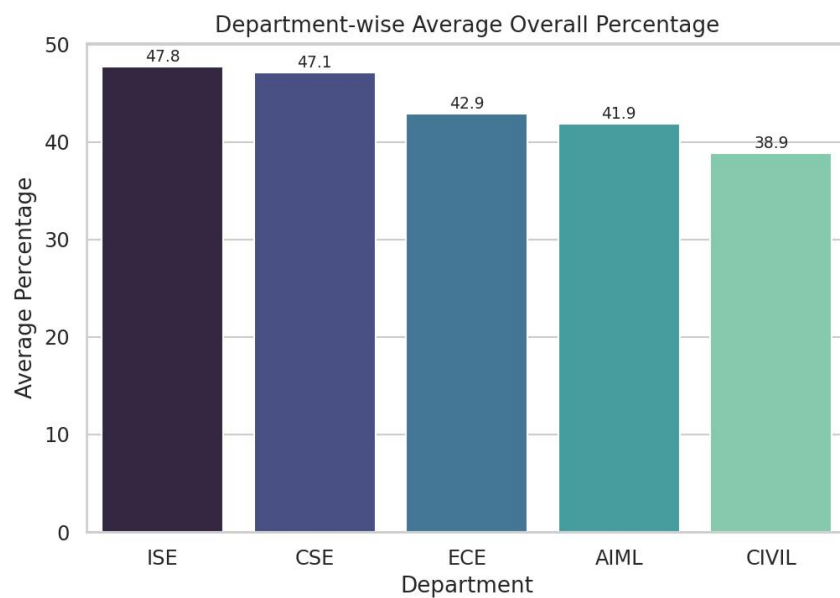


Figure 10.4 — Average overall percentage by department

ISE (47.78%) and CSE (47.12%) departments show the highest average performance, while CIVIL (38.89%) shows the lowest, a pattern that is mirrored in the department-wise pass percentages reported in Section 8.3.5.

10.5 Semester-wise Average Marks

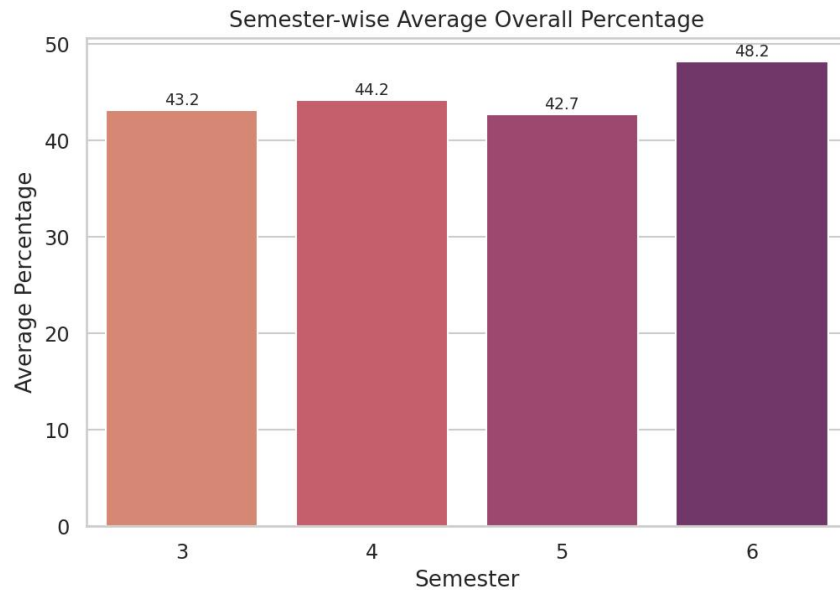


Figure 10.5 — Average overall percentage by semester

Semester 6 students show the highest average performance (48.22%), while Semester 5 shows the lowest (42.70%), though the differences across semesters are modest compared to the differences seen across attendance and study-hour bands.

10.6 Pass / Fail Distribution

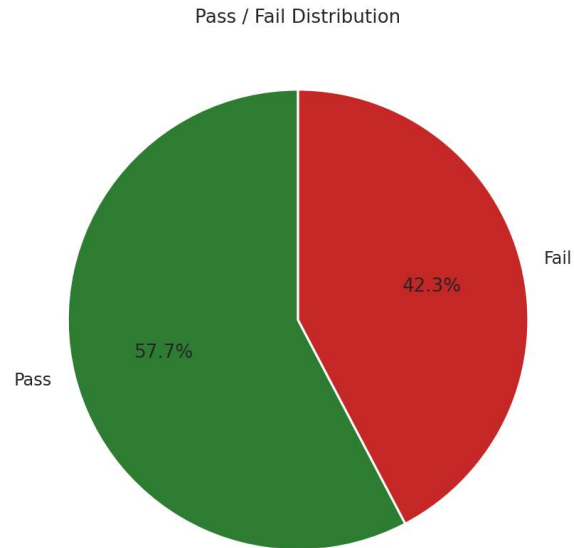


Figure 10.6 — Overall Pass/Fail distribution across the cohort

57.7% of students passed (Overall_Percentage $\geq$ 40%) while 42.3% did not meet the passing threshold, highlighting a sizeable at-risk population that could benefit from the kind of early-identification approach outlined in Chapter 11.

10.7 Correlation Heatmap

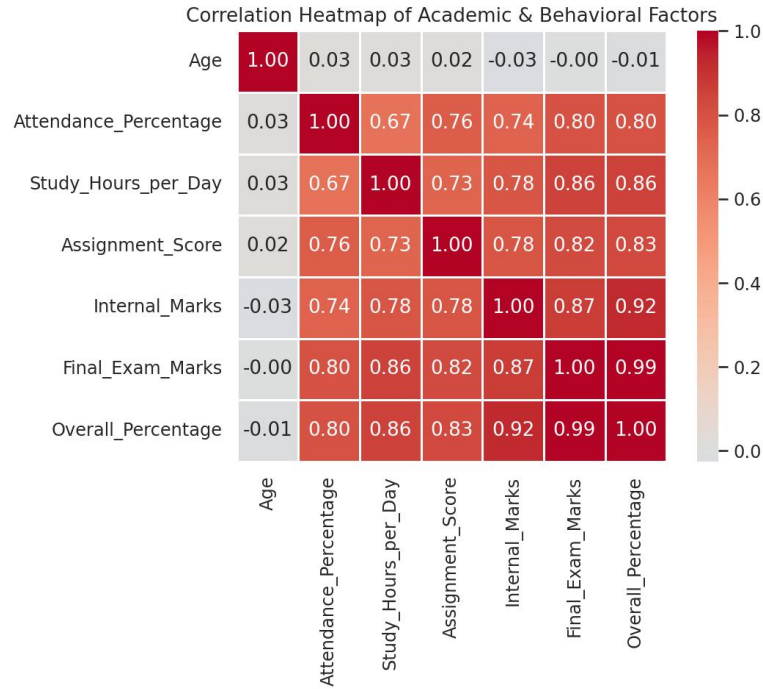


Figure 10.7 — Correlation heatmap across academic and behavioral variables

The heatmap visually confirms the findings from Section 9.2: Attendance_Percentage, Study_Hours_per_Day, Assignment_Score, Internal_Marks and Final_Exam_Marks are all strongly and positively correlated with each other and with Overall_Percentage, while Age is essentially uncorrelated with any academic variable.

10.8 Assessment-wise Performance Comparison

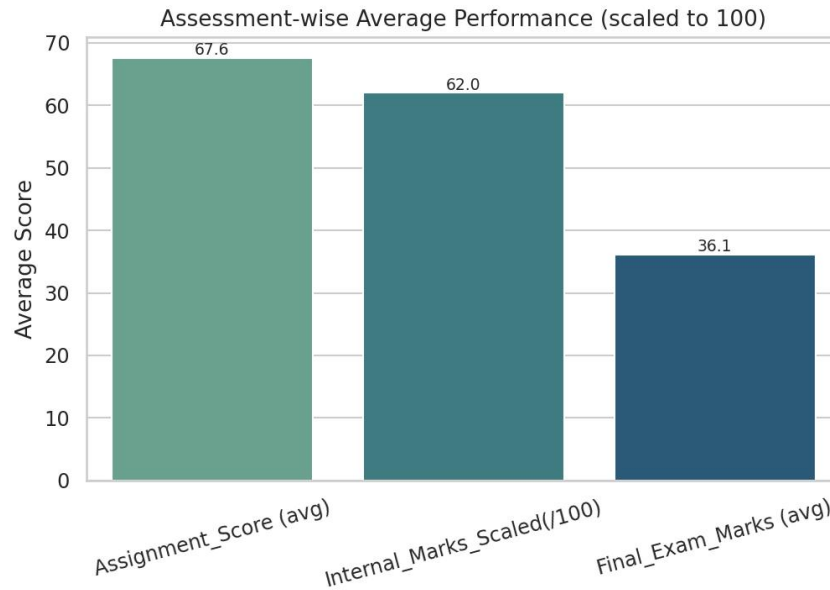


Figure 10.8 — Average performance across Assignment, Internal and Final assessments (scaled to 100)

When scaled to a common 100-point basis, Assignment_Score (67.6 average) is noticeably higher than Internal_Marks (62.0 scaled average) and Final_Exam_Marks (36.1 average), suggesting that, on average, students perform relatively better in continuous/assignment-based assessment than in the final examination — a pattern that may reflect exam pressure, time constraints, or the cumulative nature of the final exam syllabus.

11. Results and Discussion

Bringing together the SQL query outputs (Chapter 8) and the statistical/visual analysis (Chapters 9 and 10), the following key findings emerged from this project:

- Attendance and academic performance are strongly positively correlated ($r = 0.80$). Students with attendance below 75% average only 34.14% overall, compared to 64.06% for students at or above 75% attendance.
- Study hours show the strongest behavioral correlation with performance ($r = 0.86$). Average overall percentage rises steadily from 22.49% for students studying under 2 hours/day to 82.07% for those studying 6+ hours/day.
- Internal assessment marks are a strong predictor of final exam performance ($r = 0.87$ between Internal_Marks and Final_Exam_Marks), suggesting that consistent performance in continuous assessment is a reliable early indicator of final outcomes.
- Department-level differences exist but are secondary to individual behavioral factors — the gap between the best-performing department (ISE, 47.78%) and weakest (CIVIL, 38.89%) is far smaller than the gap between high- and low-attendance or high- and low-study-hour groups.
- 42.29% of students in the dataset did not meet the passing threshold (Overall_Percentage < 40%), and every one of the ten lowest-scoring students also had attendance below 66%, reinforcing attendance as a practical early-warning signal.
- Gender shows only a modest difference in average performance (Female: 46.30% vs Male: 43.56%), suggesting gender is a much weaker factor compared to attendance and study hours.
- Age shows no meaningful relationship with performance ($r = -0.01$) within the 19-22 age range represented in the dataset, which is expected given the narrow age band of a typical undergraduate cohort.

Taken together, these results suggest that attendance and daily study hours are the two most actionable behavioral levers associated with academic performance in this dataset. Because both are measurable early in a semester (unlike final exam marks, which are only available at the end), they represent practical, data-driven indicators that an institution could monitor through simple SQL queries — such as the attendance-band and study-hour-band queries developed in Chapter 8 — to flag at-risk students well before final results are declared.

12. Conclusion and Future Scope

12.1 Conclusion

This internship project successfully demonstrated an end-to-end SQL and data analytics workflow for analyzing student academic performance. Starting from a raw, imperfect dataset, the project applied systematic data cleaning, structured the data into a SQL table, and used a comprehensive set of SQL queries to answer specific academic questions — from simple aggregate statistics to department-wise pass percentages and identification of top-performing and at-risk students. This was complemented by a thorough data analytics phase using Python, in which statistical summaries, correlation analysis, and eight distinct visualizations were used to interpret and communicate the patterns found in the data.

The analysis confirmed that attendance and study hours are strongly and positively associated with academic performance, that internal assessment performance is a reliable early indicator of final outcomes, and that a substantial portion of the student cohort (42.29%) is at risk of failing — a group that can be identified early using simple, reproducible SQL queries rather than complex predictive modeling. In line with the internship's stated requirement, the project deliberately remained within the scope of SQL and Data Analytics, avoiding machine learning model-building, while still producing findings with clear, practical relevance.

12.2 Future Scope

- Extending the SQL schema to a full relational design (separate tables for students, courses, and assessments) to better reflect a real institutional database.
- Building an interactive dashboard (e.g., using Power BI, Tableau, or a Python dashboarding library) on top of the existing SQL queries for real-time monitoring by faculty.
- Incorporating additional semesters of data to study performance trends over time for the same students (longitudinal analysis).
- Once a larger, validated historical dataset is available, the analytics foundation built in this project could serve as the basis for a future, separately-scoped predictive modeling initiative — outside the boundary of the current internship.
- Automating the attendance-band and study-hour-band SQL queries developed in Chapter 8 into a scheduled report that flags at-risk students to academic advisors each month.

GIT HUB:

<https://github.com/sindhuhosagoudar725-cyber>

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